

Fixed Timer vs. AI Signal

An observation sheet for the AI Traffic Simulation activity



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K-5

MIDDLE
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Name: _____ Date: _____

Part 1 — Before You Start

Setup

Open the **AI Traffic Simulation** on the website. You will compare **Mode A (Fixed Timer)** and **Mode C (AI Adaptive)** at the same traffic volume setting. Set the Traffic Volume slider to the middle (50%) before you begin both runs.

Part 2 — Run Mode A: Fixed Timer

Click Mode A, press Run, and let it play for about 30 seconds. Then record:

Question	Your Observation
How many vehicles were processed?	
What was the average wait time?	
What was the congestion level (%)?	
Did any lane look empty while its light was still green? (Yes / No)	

Part 3 — Run Mode C: AI Adaptive

Reset, switch to Mode C, press Run, and let it play for about 30 seconds. Record:

Question	Your Observation
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How many vehicles were processed?	
What was the average wait time?	
What was the congestion level (%)?	
Did the AI seem to switch lights more often or less often?	

Part 4 — Compare & Reflect

1. Which mode processed more vehicles in the same amount of time?

2. Which mode had a shorter average wait time?

3. Why do you think the AI mode performed differently than the fixed timer? Use the word **'queue'** or **'sensor'** in your answer.

4. Can you think of a real place near your home where smarter traffic signals might help? Describe it.

For Teachers

This activity reinforces the engineering design concept of feedback loops — students should notice the AI mode reacts to real conditions while the fixed timer does not.

